

ITI0201

**ROBOTS PROGRAMMING
(EUROTEQ CATALOG COURSE)**

TWEEDBACK

[HTTPS://TWEEDBACK.DE/UJCX](https://tweedback.de/UJCX)



WHY THIS COURSE?

- Robotics is a rapidly evolving field that is growing blazingly fast
- The bottleneck in robotics is in software (not hardware)

WHO IS THE INSTRUCTOR?

- Senior lecturer Dr. Gert Kanter
 - Doctor of Philosophy (Computer Science)
- Research mostly in the following areas
 - robotics software (e.g., software engineering for robotics, algorithms)
 - autonomous robots software testing
 - full-stack robotics development
- Currently working on humanoid robots

PHILOSOPHY OF THE COURSE

- "Hardcore" programming course
- Personalized feedback
- Studying with fellow students (teamwork)

COURSE ORGANIZATION

MOODLE

- Moodle aine "ITI0201 Robots Programming (2025)"
- Moodle link:
<https://moodle.taltech.ee/course/view.php?id=34577>
 - No enrolment key

SCHEDULE

- Lecture (consultation sessions)
 - Odd-numbered weeks (8 sessions total)
- Labs
 - Odd-numbered weeks 1.5h ($8 * 1.5h$)
 - Even-numbered weeks ($1.5h + 1.5h$) = 3h ($8 * 3h$)

CONTENT ORGANIZATION

The course content is divided into four **tracks** based on the study method and scope.

TRACKS

- **Track 1:** Focused exercises with automated tests.
- **Track 2:** Simulation-based programming of ideal noise-free robots.
- **Track 3:** Programming of non-ideal robots (realistic simulation or real-world robot).
- **Track 4:** Weekly reflection on personal study process and feedback.

TEAM-BASED LEARNING

- T1-T3 tasks are designed to be solved in **teams of two**.
- T4 reflection and feedback tasks are individual.

TEAMS

- Teams are self-formed.
- Each team should have **different teammates** for every major task (i.e., shapes, colors, and maze).
 - Promotes diverse teamwork experience and collaboration with new people.
 - 3 teams over the semester.
- After finishing the major task, teammates will be evaluated and given feedback by their partners.
 - Feedback makes it possible to improve.

T1 (TRACK 1)

- Tasks **EX01 to EX10** are equipped with automated tests.
- Worth approximately **23%** of total points.

T2 (TRACK 2)

- Simulation-based programming of ideal, noise-free robots.
- Worth approximately **57%** of total points.

T3 (TRACK 3)

- Programming of non-ideal robots (realistic simulation or real-world robot).
- Worth approximately **18%** of total points.

T4 (TRACK 4)

- Self-reflection every week.
- The goal is to help you identify problems and improve your personal and overall learning process.
- Also serves as a feedback system for the teaching team.
- As an extra incentive, every reflection awards 1 point (0.1% of total points, 1.6% for 16 weeks).
- Teammate feedback for each major task for improving teamwork.
- One overall course reflection at the end of the semester.

TRACKS OVERVIEW

- **T1:** 22.9% (10 tasks, total 235 points).
- **T2:** 56.6% (main focus, total 580 points).
- **T3:** 18.1% (extensions to T3, total 185 points).
- **T4:** 2.4% (16 tasks $\times$ 1 points + 3 $\times$ 2 points + 3 points = 24 points).
- Feedback + reflection: maximum $15 \times 2 + 9 = 39$ points.

Total: $24 + 235 + 580 + 185 = 1024$ points.

T3: REAL-WORLD CHALLENGES

- Real-world T3 tasks are often harder but also fully "realistic."

WHY DOES T3 EARN FEWER POINTS DESPITE BEING HARDER?

1. **Lab time limitations** prevent allocating a higher weight to T3 in credit calculations.
2. The course is designed for computer science students, with an emphasis on programming solutions rather than hardware related problem-solving.
3. Once T2 tasks are solved with correct techniques, T3 tasks require less time (especially in simulation).

LAB TIME BREAKDOWN FOR T3

$$\frac{((1.5 + 3)/2) \times 16}{6 \times 26} = 0.2308 \approx 23\%$$

Lab time also includes defending the tasks, leaving even less time for testing.

PROCESS

T1 → T2 → T3 → T4

- **T1:** Develops required components.
- **T2:** Applies T1 components to larger tasks.
- **T3:** Adds noise filters and other features to T2 tasks.
- **T4:** Aids knowledge consolidation and improves the learning process.

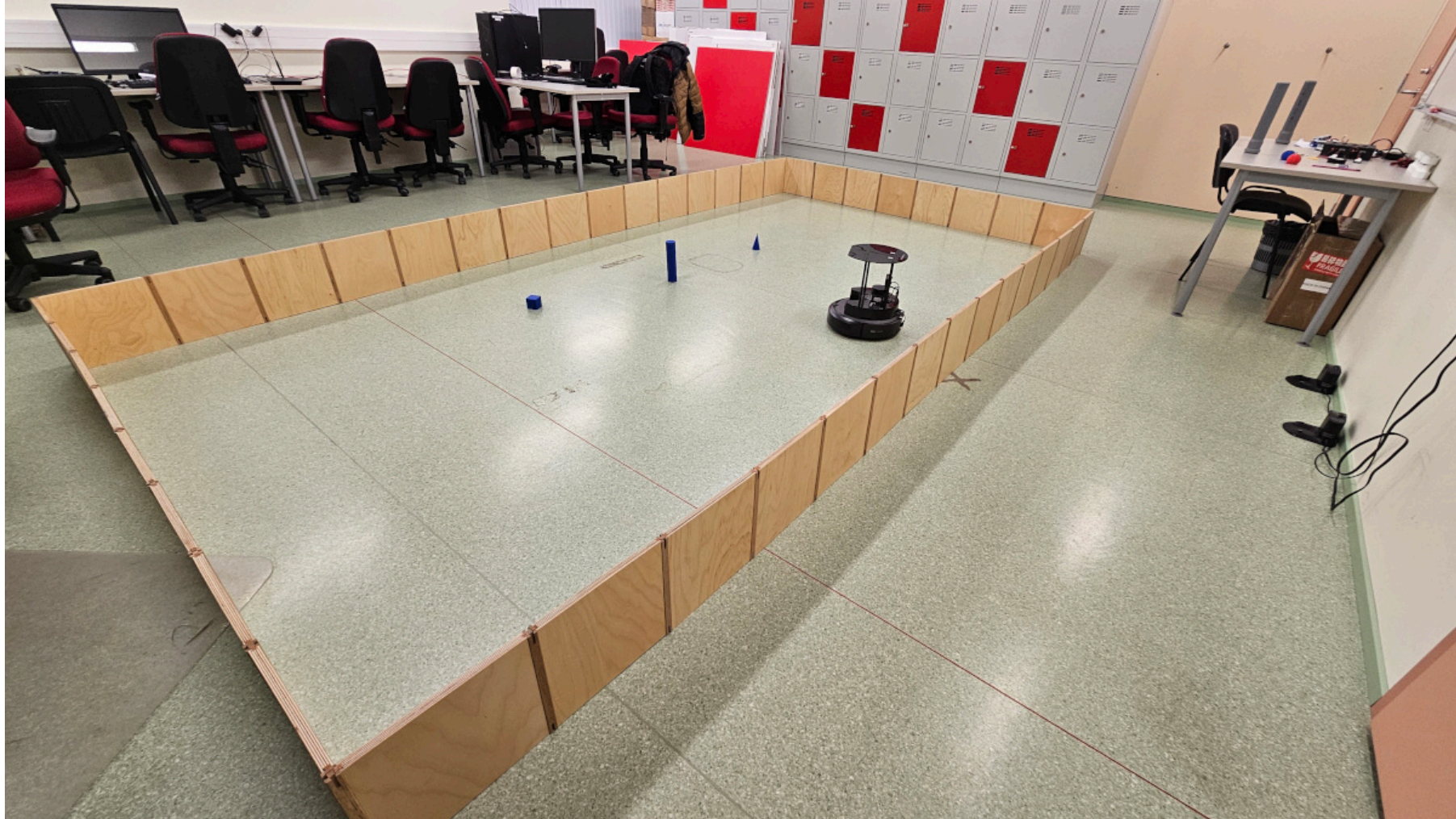
ASSESSMENT

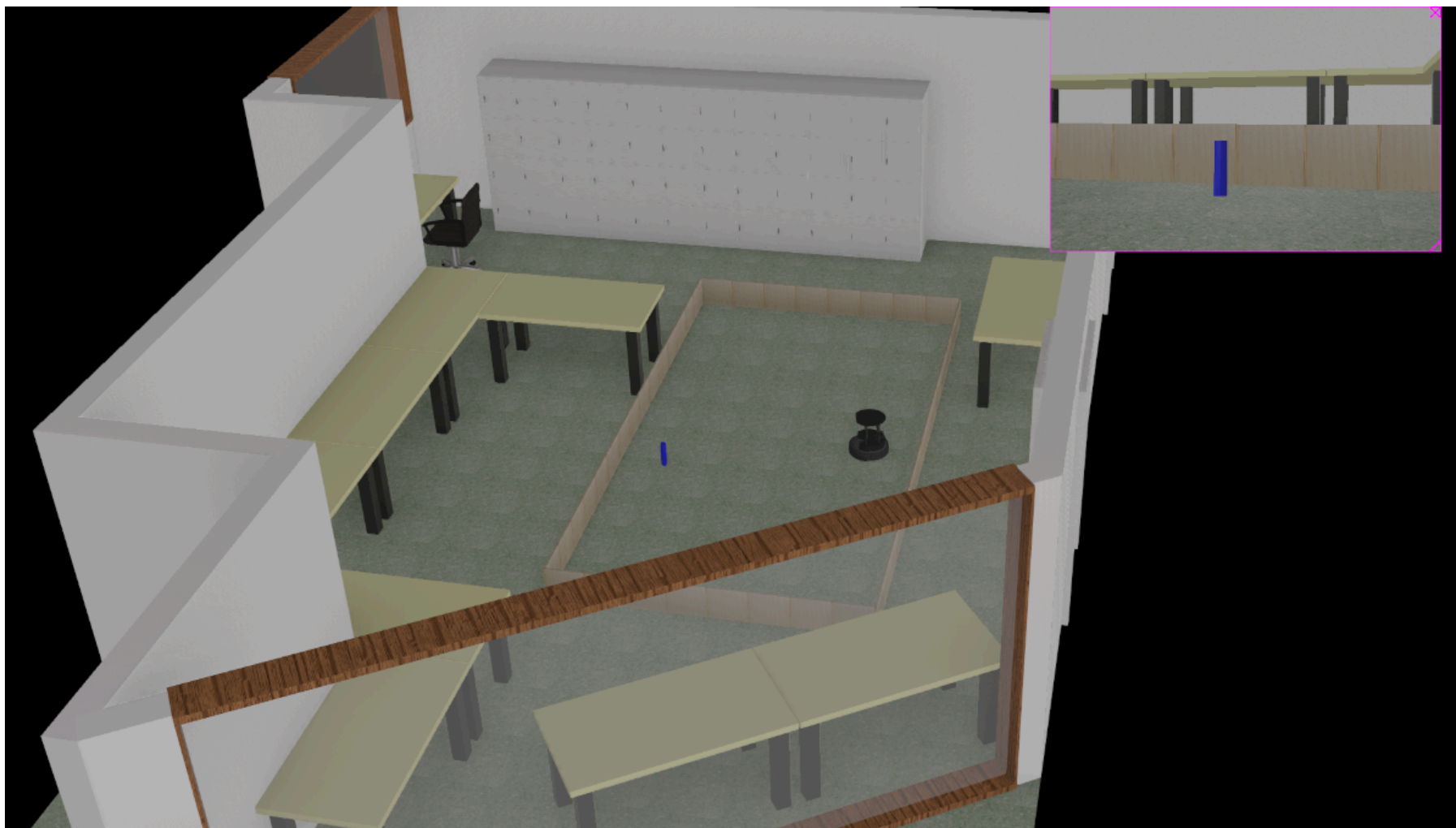
A traditional point-based system.

GRADING SCALE

- **5 (Excellent):** 90.1%+ (901+ points).
- **4 (Very Good):** 80.1%–90.0% (801–900 points).
- **3 (Good):** 70.1%–80.0% (701–800 points).
- **2 (Satisfactory):** 60.1%–70.0% (601–700 points).
- **1 (Poor):** 50.1%–60.0% (501–600 points).
- **0 (Fail):** Below 50.1% (<501 points).

TURTLEBOT4 ROBOT





T2-T3 TASKS

1. **Shapes (S):** 4 weeks.
2. **Colors (C):** 5 weeks.
3. **Maze (M):** 6 weeks.

SHAPES

- **Level 1:** The robot detects an object and moves toward it.
- **Level 2:** The robot iteratively detects objects and alternates between approaching the closest and the farthest object.
- **Level 3:** The robot positions itself to form an equilateral triangle with two objects.

COLORS

- **Level 1:** The robot finds the blue object and moves toward it.
- **Level 2:** The robot iteratively detects objects and alternates between approaching the closest and the farthest blue object.
- **Level 3:** The robot detects the blue cube (ignores other shapes such as cone and cylinder) and moves toward it.

MAZE

- **Level 1:** Robot exits a maze consisting of U-turns.
- **Level 2:** Robot exits a random maze.
- **Level 3:** Robot fully maps the maze and exits it.
- **Level 4:** Robot fully maps the maze, is moved to a random place, and moves to a pre-determined corner.

TASK DEFENSE

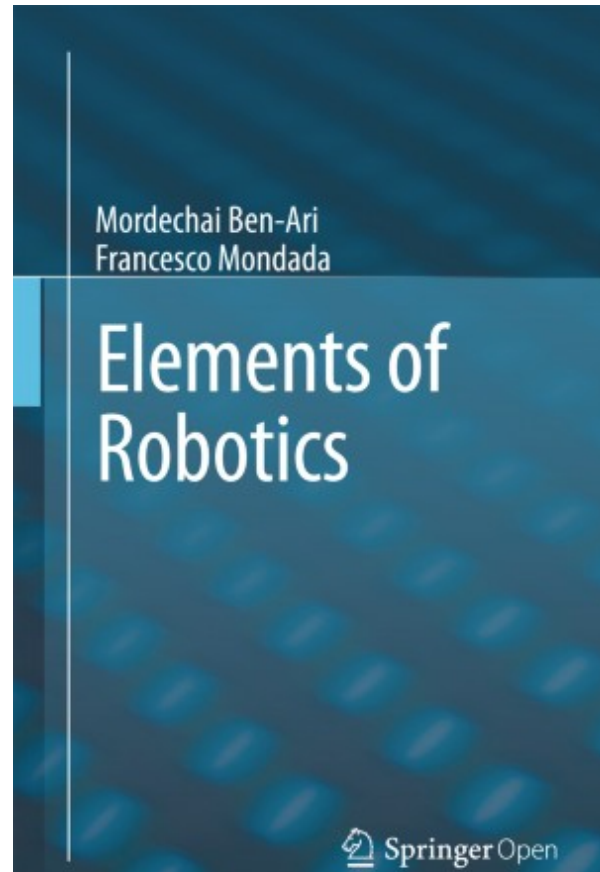
- T1–T3 tasks must be presented and defended in teams.
- Both team members must be present.

LAB SESSIONS

- Individual feedback is crucial!
- Be active and ask questions!

RECOMMENDED READING FOR BEGINNERS

Open Access book: *Elements of Robotics*



NEXT STEPS

THIS WEEK

- Select a teammate for the Shapes (S) task.
- **T1:** Solve EX01 and EX02 with your teammate.
- **T2-T3:** No graded tasks yet; We are focusing on setting up the simulation environment during labs.
- **T4:** Reflect on your studies after the first week.